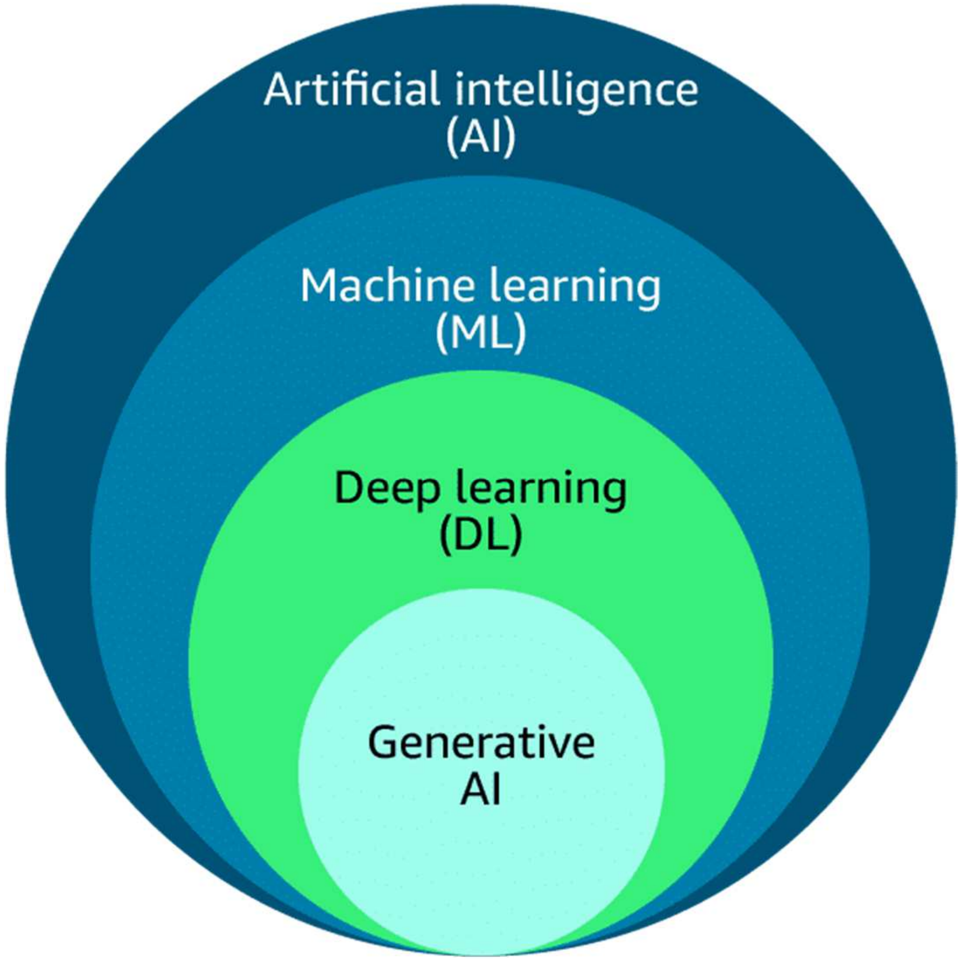
Fundamentals of Machine Learning and Artificial Intelligence

Introduction

You will explore the connections between AI, ML, deep learning, and the emerging field of generative artificial intelligence (generative AI), which has captured the attention of businesses and individuals alike. You will gain a solid understanding of foundational AI terms, laying the groundwork for a deeper dive into these concepts. Additionally, you will learn about a selection of Amazon Web Services (AWS) services that use AI and ML capabilities.

Artificial Intelligence

Generative AI
Generative AI is a subset of deep learning because it can adapt models built using deep learning, but without retraining or fine tuning. Generative AI systems are capable of generating new data based on the patterns and structures learned from training data.



Machine Learning Fundamentals

Labeled data is a dataset where each instance or example is accompanied by a label or target variable that represents the desired output or classification. These labels are typically provided by human experts or obtained through a reliable process.

Example: In an image classification task, labeled data would consist of images along with their corresponding class labels (for example, *cat*, *dog*, *car*).

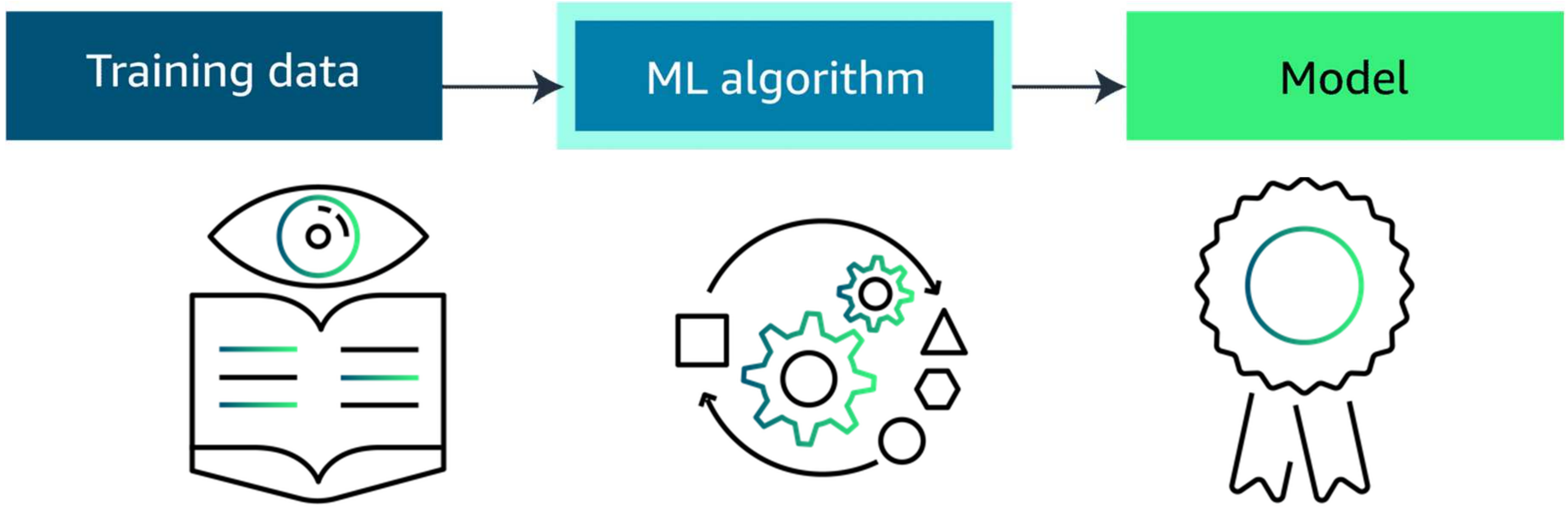
Unlabeled data is a dataset where the instances or examples do not have any associated labels or target variables. The data consists only of input features, without any corresponding output or classification.

Example: A collection of images without any labels or annotations

Types of Data

Structured Data	Unstructured Data
Structured data refers to data that is organized and formatted in a predefined manner, typically in the form of tables or databases with rows and columns. This type of data is suitable for traditional machine learning algorithms that require well-defined features and labels. The following are types of structured data.	Unstructured data is data that lacks a predefined structure or format, such as text, images, audio, and video. This type of data requires more advanced machine learning techniques to extract meaningful patterns and insights.
Tabular data: This includes data stored in spreadsheets, databases, or CSV files, with rows representing instances and columns representing features or attributes. **Time-series data:** This type of data consists of sequences of values measured at successive points in time, such as stock prices, sensor readings, or weather data.	**Text data:** This includes documents, articles, social media posts, and other textual data. **Image data:** This includes digital images, photographs, and video frames.

ML Algorithm



In *supervised learning*, the algorithms are trained on labeled data. The goal is to learn a mapping function that can predict the output for new, unseen input data.

Unsupervised learning refers to algorithms that learn from unlabeled data. The goal is to discover inherent patterns, structures, or relationships within the input data.

In *reinforcement learning*, the machine is given only a performance score as guidance and semi-supervised learning, where only a portion of training data is labeled. Feedback is provided in the form of rewards or penalties for its actions, and the machine learns from this feedback to improve its decision-making over time.

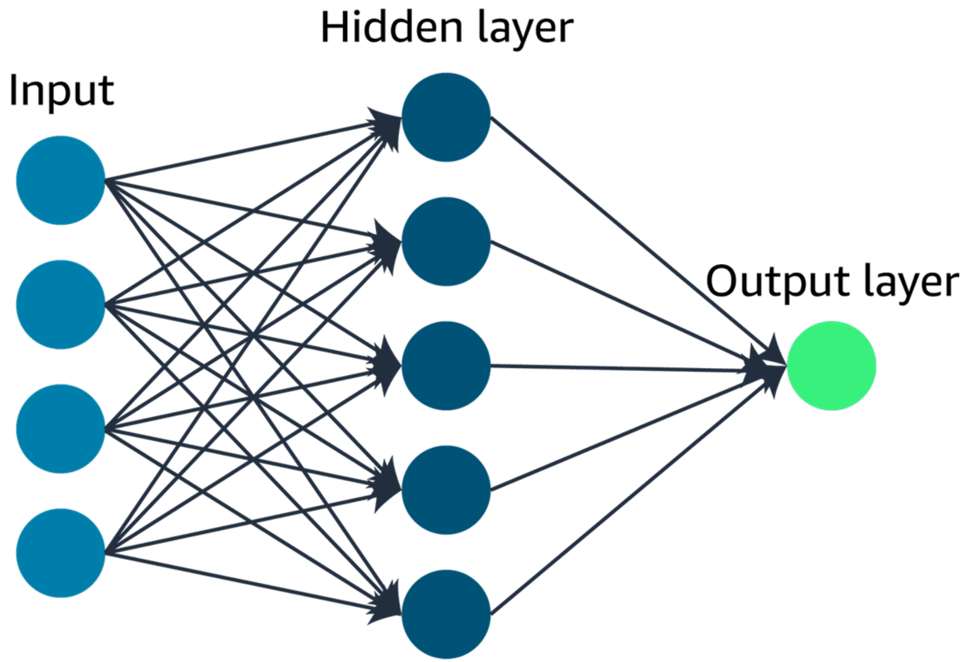
Inferencing

Process of using the information that a model has learned to make predictions or decisions.
This is called **inferencing.**

Batch Inferencing
This type of inferencing is often used for tasks like data analysis, where the speed of the decision-making process is not as crucial as the accuracy of the results. Normally involves large datasets.

Real-time Inferencing
This is important for applications where immediate decision-making is critical, such as in chatbots or self-driving cars. The computer has to process the incoming data and make a decision almost instantaneously, without taking the time to analyze a large dataset.

Deep Learning Fundamentals



- Inspired by the structure and function of the human brain. Use of artificial neural networks, which are computational models designed to mimic the human brain processes information.

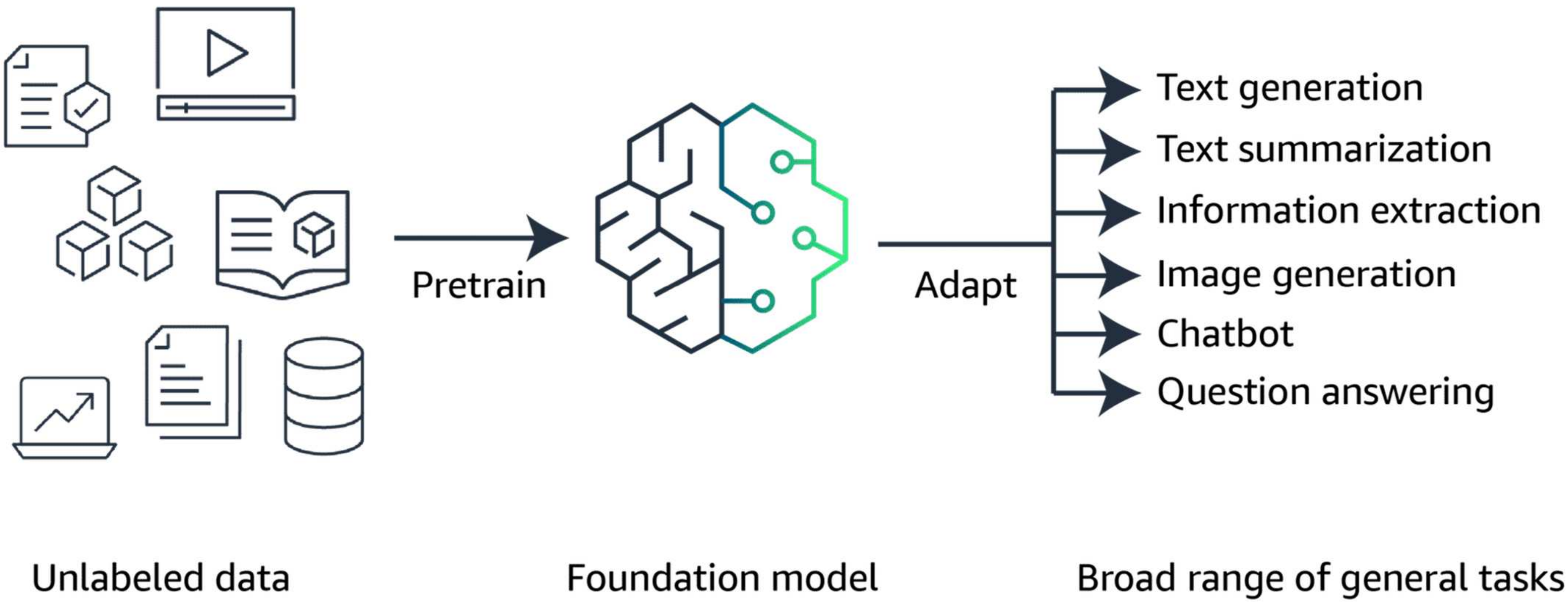
- Neural network learns to recognize patterns from examples, it can then look at data for completely new customers that it has never seen before and still make predictions about what they might buy or how they might behave for example.

Use of Deep Learning to Enhance Results

- **Computer vision** is a field of artificial intelligence that makes it possible for computers to interpret and understand digital images and videos.
  Deep learning has revolutionized computer vision by providing powerful techniques for tasks such as image classification, object detection, and image segmentation.

- **Natural language processing (NLP)** is a branch of artificial intelligence that deals with the interaction between computers and human languages.
  Deep learning has made significant strides in NLP, making possible tasks such as text classification, sentiment analysis, machine translation, and language generation.

Generative AI



Generative AI and Foundation Models

Foundation Models (FMs)

- Generative AI uses foundation models (FMs) pretrained on large-scale internet data.
- A single FM can be adapted to handle multiple tasks like text generation, summarization, image creation, and chatbots.
- FMs eliminate the need for separate labeled datasets and serve as a base for specialized models.

Large Language Models (LLMs)

- LLMs are commonly built on transformer architectures for processing and generating human-like text.
- They are trained on massive text datasets from diverse sources such as books and the internet.
- LLMs learn complex patterns and relationships in language, enabling advanced text understanding and generation.

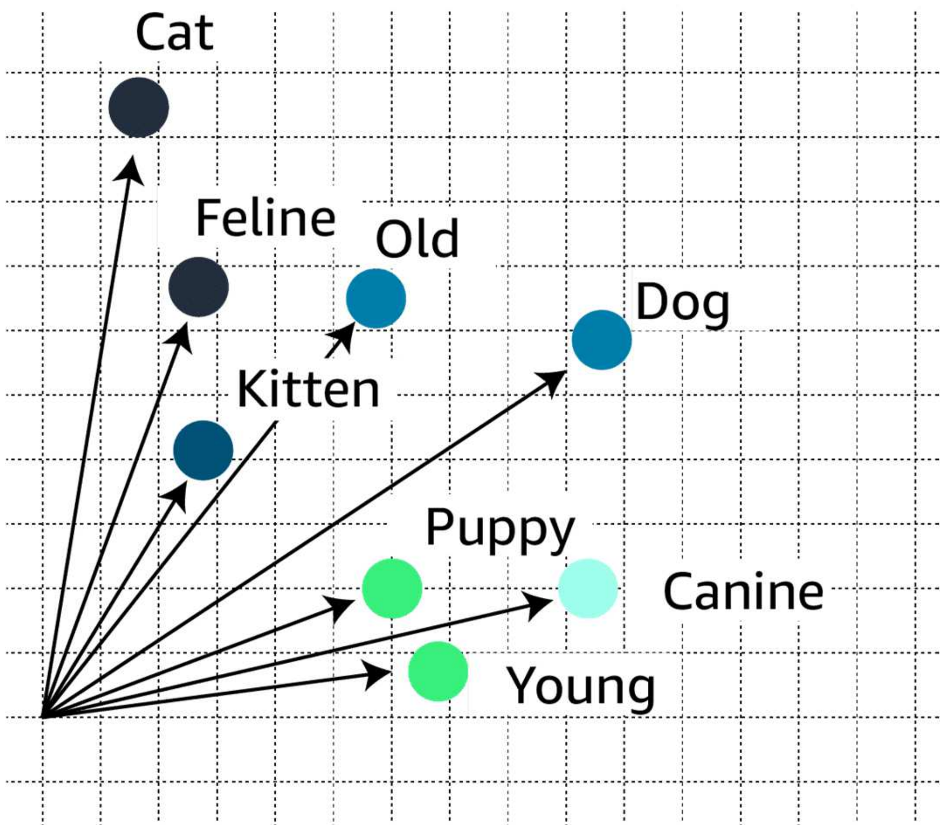
Tokens, Embeddings and Vectors

Tokens are the basic units of text that the model processes. Tokens can be words, phrases, or individual characters like a period. Tokens also provide standardization of input data, which makes it easier for the model to process.

Tokens, Embeddings and Vectors

Embeddings are numerical representations of tokens, where each token is assigned a vector (a list of numbers) that captures its meaning and relationships with other tokens.

These **vectors** are learned during the training process and allow the model to understand the context and nuances of language.



Diffusion Models

Diffusion Models:

- Diffusion is a deep learning architecture that begins with pure noise and progressively refines it into a clear output like an image or text.
- The model adds meaningful information step-by-step to transform the initial random data into coherent results.
- Diffusion models learn using a two-step process: forward diffusion (adding noise) and reverse diffusion (removing noise to generate output).

Forward diffusion

Using forward diffusion, the system gradually introduces a small amount of noise to an input image until only the noise is left over.

Reverse diffusion

In the subsequent reverse diffusion step, the noisy image is gradually introduced to denoising until a new image is generated.

Multimodal Models

- Multimodal models can process and generate multiple types of data (e.g., text, images) simultaneously instead of relying on a single input/output type.
- They learn relationships between different modalities, like how text and images are connected and influence each other.
- Applications include video captioning, graphic creation from text, intelligent question answering and content translation with visual preservation.

Other generative models are as such:

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)

Optimizing Model Outputs

- **FM** can be further optimized in several different ways. Techniques vary in complexity and cost, with the fastest and lowest cost option being prompt engineering.
- Prompt Engineering act as instructions for Gen AI FM models.
  - **Instructions**: This is a task for the FM to do. It provides a task description or instruction for how the model should perform.
  - **Context**: This is external information to guide the model.
  - **Input data**: This is the input for which you want a response.
  - **Output indicator**: This is the output type or format.

Example prompt

You are an experienced journalist that excels at condensing long articles into concise summaries. Summarize the following text in 2–3 sentences.

Text: [Long article text goes here]

Fine Tuning

- Fine-tuning FM base model can improve performance.
- Is a supervised learning process involves taking a pre-trained model and adding specific, smaller datasets.
- Adding narrower datasets modifies the weights of the data to better align with the task.

Two ways to fine-tune a model:
Instruction fine-tuning uses examples of how the model should respond to a specific instruction. Prompt tuning is a type of instruction fine-tuning.

Reinforcement learning from human feedback (RLHF) provides human feedback data, resulting in a model that is better aligned with human preferences.

Retrieval-Augmented Generation (RAG)

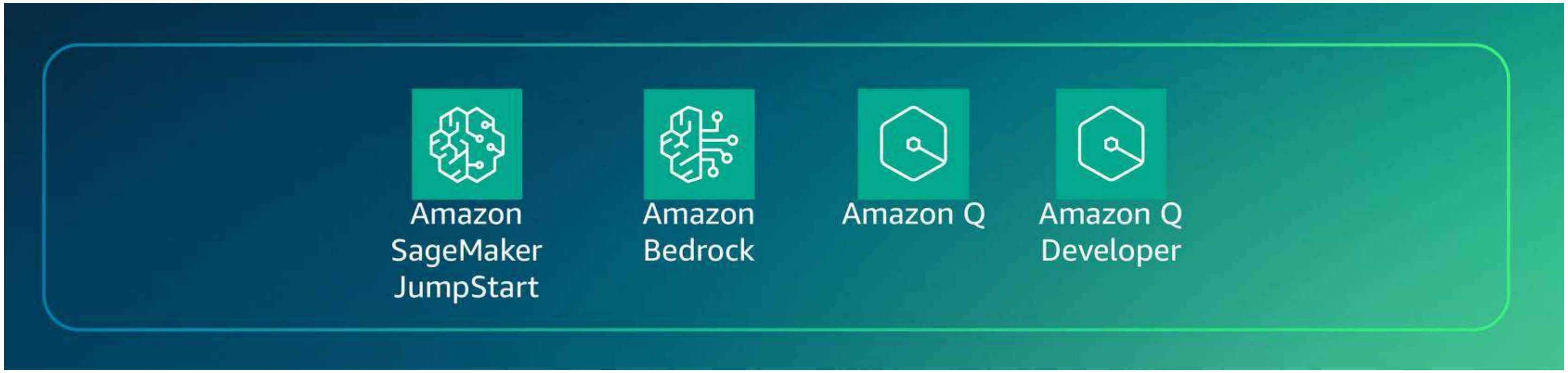
- RAG provides domain-specific data as context to generate more accurate and relevant responses without retraining the model.
- Unlike fine-tuning, RAG retrieves relevant documents to guide answers instead of adjusting the foundation model's weights.
- Fine-tuning alters model weights with labeled examples, while RAG keeps the original model unchanged and supplements it with external information.

AWS Infrastructure and Technologies

ML frameworks layer plays a crucial role in the development and deployment of machine learning models. At the core of the frameworks layer is **Amazon SageMaker**. SageMaker offers the right tools to effectively build, train, and run LLMs and other FMs efficiently and cost effectively.

AI/ML Services

Generative AI



Generative AI Services Layer:

- This layer in the AI/ML stack provides specialized tools for generative AI, such as SageMaker JumpStart for quick model development and deployment.
- Amazon Bedrock enables access to top-performing foundation models via a single API, offering flexibility and ease of use.
- These services empower developers to build solutions for content creation, data synthesis, and interactive AI applications.

Generative AI

SageMaker JumpStart:

- SageMaker JumpStart offers ready-to-deploy solutions for common ML use cases, helping users begin their ML journey quickly.
- It includes customizable solutions using AWS CloudFormation templates and reference architectures.
- Supports one-click deployment and fine-tuning of 150+ popular open-source models across NLP, object detection, and image classification

Amazon Bedrock:

- A fully managed, serverless service that provides access to foundation models (FMs) from Amazon and leading AI startups via a unified API.
- Enables rapid experimentation, private customization with your data, and easy integration of FMs into AWS applications.

Generative AI

Amazon Q:

- A generative AI assistant that uses your organization's data, code, and enterprise systems to provide accurate, context-aware answers and actions.
- Helps streamline tasks, accelerate decisions, and boost creativity and innovation through fast, relevant insights.

Amazon Q Developer:

- Boosts developer productivity with ML-powered code recommendations for languages like C#, Java, JavaScript, Python, and TypeScript.
- Integrates with multiple IDEs to generate entire functions or code blocks, accelerating application development.

Cost Considerations

Understand the various **cost considerations** involved.
These can impact factors such as responsiveness, availability, redundancy, performance, regional coverage, pricing models, throughput and the ability to use custom models.

Responsiveness and availability	Redundancy and Regional coverage	Performance
Token-based pricing	Provisioned throughput	Custom models

Knowledge Check
Q1.

A company wants to develop a system that can accurately recognize and classify handwritten digits from images.
Which of the following options best describes the use of neural networks for this task?

A. Neural networks are a type of decision tree algorithm that can be trained on image data to create a set of rules for classifying handwritten digits.

B. Neural networks are a form of linear regression that can be used to map pixel values from images to corresponding digit labels.

C. Neural networks are a type of deep learning model inspired by the structure and function of the human brain. They consist of interconnected nodes that can learn to recognize patterns in data, such as images of handwritten digits.

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Q2.

A company is developing an artificial intelligence (AI) system to control a self-driving car. The system learns through trial-and-error interactions with the driving environment, receiving rewards for safe and efficient actions.

Which machine learning (ML) approach is being used in this scenario?

A. Supervised learning

B. Reinforcement learning

C. Unsupervised learning

D. Self-supervised learning

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Q3.

A company is developing a large language model (LLM) for natural language processing tasks, such as text generation, summarization, and question answering.
Which of the following best describes the role of embeddings, in the context of LLMs?

A. Embeddings are numerical representations of words or tokens, where semantically similar words have similar vector representations.

B. Embeddings are the preprocessing techniques used to clean and tokenize the text data before feeding it into the LLM for training or inference.

C. Embeddings are the ensemble methods used to combine multiple LLMs to improve the overall performance and robustness of the system.

D. Embeddings are the linguistic rules and grammar patterns extracted from the text data to aid the LLM in understanding and generating language.

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Q4.

A company has pre-trained a large language model on a vast corpus of text data. They want to adapt this pre-trained model to perform specific tasks such as sentiment analysis and document summarization.

Which of the following best describes the process of fine-tuning?

A. Fine-tuning involves training the pre-trained language model from scratch.

B. Fine-tuning refers to the process of further training the pre-trained language model on labeled data for the specific tasks.

C. Fine-tuning is a technique used to preprocess and clean the task-specific data before feeding it into the pre-trained language model.

D. Fine-tuning is an ensemble method that combines the pre-trained language model with task-specific models to improve the overall performance.

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- A company has a large collection of customer support emails and chat transcripts. They want to analyze the sentiment expressed in these messages and identify common issues or topics discussed by their customers.
  Which AWS service would be most appropriate for this task?

A. Amazon Transcribe

B. Amazon Kendra

C. Amazon Polly

D. Amazon Comprehend

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A retail company has accumulated a large volume of customer transaction data, including purchase history, product preferences, and demographic information. The company wants to use this data to build machine learning models that can provide personalized product recommendations to customers and improve their overall shopping experience.

Which AWS service would be most suitable for the retail company to build, train, and deploy machine learning models for personalized product recommendations?

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